Status	Finished
Started	Tuesday, 20 October 2026, 6:02 PM
Completed	Tuesday, 20 October 2026, 6:05 PM
Duration	3 mins 17 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By sorting the list first
c. By dividing the list into halves
d. By comparing each element in the list sequentially ✓

Question **2**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The middle element is chosen randomly
b. The lower middle element is chosen as the middle element ✗
c. The higher middle element is chosen as the middle element
d. The search stops

Question **3**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓
b. In the left sublist
c. At the beginning of the list
d. In the middle of the list

Question **4**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It can be applied only if the list is sorted ✓
- b. It compares elements sequentially
- c. It works on unsorted lists
- d. It always starts from the beginning of the list

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. There must be a mechanism to access middle element directly
- b. Must use a sorted array
- c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **6**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O👎
- b. O(1) ✓
- c. O(n log n)
- d. O(log n)

Question **7**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n/2
- b. n ✓
- c. n-1
- d. log n

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Skipping every second element
- c. Sorting the list before searching
- d. Dividing the list in half repeatedly

Question **9**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 1000
- b. 500
- c. 100
- d. 10 ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Bubble search
- b. Merge search
- c. Linear search ✓
- d. Quick search

Question **11**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. An error is raised
- b. The search is considered unsuccessful ✓
- c. The first element is returned
- d. The last element is returned

Question **12**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Hash search

b. Linear search ✓

c. Both (1) & (3)

d. Binary search

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues in the left or right sublist ✓

b. The search stops

c. The search continues from the beginning

d. The list is sorted

Question **14**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Insertion

c. Complexity

d. Bubble ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O(log n) ✓
c. O(n log n)
d. O👎